1 **(a)** 0.17 **[B1]**

(b) The train does not arrive on time (or equivalent) **[B1]**

[Total Marks: 2]

2 **(a)** $0.15 + 2x + 0.31 + x + 0.24 = 1$ or $3x + 0.7 = 1$ **[M1]**

0.1 **[A1]**

(b) $P(2) = 2(0.1) = 0.2$ **[M1]**

0.8 **[A1]**

[Total Marks: 4]

3 **(a)** $\frac{1}{5} + 0.35 + x + 2x = 1$ or $0.55 + 3x = 1$ **[M1]**

0.15 **[A1]**

(b) $1 - 2(0.15)$ or $\frac{1}{5} + 0.35 + 0.15$ **[M1]**

0.7 or $\frac{7}{10}$ **[A1]**

[Total Marks: 4]

4 **(a)** $x^2 + 3x + 2$ **[B1]**

(b) $\frac{x^2}{x^2 + 3x + 2} = \frac{8}{15}$ **[M1]**

$15x^2 = 8(x^2 + 3x + 2) \implies 15x^2 = 8x^2 + 24x + 16$ **[M1]**

(c) $(7x + 4)(x - 4) = 0$ or correct substitution into the quadratic formula [M2]
(Award M1 for $(7x + a)(x + b) = 0$ where $ab = -16$ or $7b + a = -24$)

4 **[A1]**

(d) Total counters $= 4^2 + 3(4) + 2 = 30$ and number of blue counters $= 3(4) = 12$ **[M1]**

$\frac{18}{30} = \frac{3}{5}$ **[A1]**

[Total Marks: 8]

5 **(a)** 21 in the intersection $C \cap D$ **[B1]**

34 in C only and 47 in D only **[B1]**

18 outside both circles **[B1]**

(b) $\frac{21}{120}$ or $\frac{7}{40}$ **[B1]**

(c) $1 - \frac{68}{120}$ or $\frac{34 + 18}{120}$ **[M1]**

$\frac{52}{120}$ or $\frac{13}{30}$ **[A1]**

(d) $\frac{34+47}{120}$ or $\frac{120-21-18}{120}$ **[M1]**

$\frac{81}{120}$ or $\frac{27}{40}$ **[A1]**

[Total Marks: 8]

6 (a) $(x+4)+x+(2x-1)+(3x+5)=50$ or $7x+8=50$ **[M1]**

6 **[A1]**

(b) $(2(6)-1)+(3(6)+5)$ or $50-(6+4+6)$ or $\frac{11+23}{50}$ **[M1]**

$\frac{34}{50}$ or $\frac{17}{25}$ or 0.68 **[A1]**

(c) $(6+4)+(2(6)-1)$ or $10+11$ or $\frac{21}{50}$ **[M1]**

$\frac{21}{50}$ or 0.42 **[A1]**

[Total Marks: 6]

7 (a) Sum of given probabilities: $0.15+0.3+0.45=0.9$ **[M1]**

0.1 **[A1]**

(b) 0.45 **[B1]**

(c) Uses $1-P(G)$ formula: $1-0.45$ **[M1]**

0.55 **[A1]**

[Total Marks: 5]

8 (a) $1-\left(\frac{2}{5}+\frac{1}{3}\right)$ or $1-\frac{11}{15}$ **[M1]**

$\frac{4}{15}$ **[A1]**

(b) Equates their fraction to the number of green marbles: $\frac{4}{15}$ of total $= 12$ **[M1]**

Total $= 12 \times \frac{15}{4}$ **[M1]**

45 **[A1]**

(c) $\frac{2}{3}$ **[B1]**

[Total Marks: 6]

9 (a) Places $2x$ in the intersection **[B1]**

Places $38 - 2x$ in T only region and $24 - 2x$ in C only region **[B1]**

Places x in the region outside both circles **[B1]**

(b) Forms correct equation: $(38 - 2x) + 2x + (24 - 2x) + x = 60$ or $62 - x = 60$ **[M1]**

2 **[A1]**

(c) Finds the number of people in $T \cap C'$: $38 - 2(2) = 34$ **[M1]**

$\frac{34}{60}$ or $\frac{17}{30}$ **[A1]**

[Total Marks: 7]